Ch. 15 CW #3

Principles of Microeconomics

December 12, 2025

Instructions

Please answer the following on a separate sheet of paper, fully justify your answers, and turn your answers in on Gradescope by the beginning of next class.

Problems and Applications

9. The market for apple pies in the city of Ectenia is competitive and has the following demand schedule:

Price	Quantity Demanded
\$1	1,200 pies
2	1,100
3	1,000
4	900
5	800
6	700
7	600
8	500
9	400
10	300
11	200
12	100
13	0

Each producer in the market has fixed costs of \$9 and the following marginal cost schedule:

Quantity	Marginal Cost
1 pie	$2
2	4
3	6
4	8
5	10
6	12

(a) Compute each producer's total cost and average total cost for each quantity from 1 to 6 pies.

(b) The price of a pie is now $11. How many pies are sold? How many pies does each producer make? How many producers are there? How much profit does each producer earn?

(c) Is the situation described in part (b) a long-run equilibrium? Why or why not?

(d) Suppose that in the long run, there is free entry and exit. How much profit does each producer earn in the long-run equilibrium? What is the market price? How many pies does each producer make? How many pies are sold in the market? How many pie producers are operating?

10. An industry currently has 100 firms, each of which has fixed costs of $16 and average variable cost as follows:

Quantity	Average Variable Cost
1	$1
2	2
3	3
4	4
5	5
6	6

(a) Compute a firm's marginal cost and average total cost for each quantity from 1 to 6.

(b) The equilibrium price is currently $10. How much does each firm produce? What is the total quantity supplied in the market?

(c) In the long run, firms can enter and exit the market, and all entrants have the same costs as above. As this market makes the transition to its long-run equilibrium, will the price rise or fall? Will the quantity demanded rise or fall? Will the quantity supplied by each firm rise or fall? Explain your answers.

(d) Graph the long-run supply curve for this market with specific numbers on the axes as relevant.

11. Suppose that each firm in a competitive industry has the following costs:

Total cost:	$TC = 50 + \frac{1}{2}q^2$
Marginal cost:	$MC = q$

where q is an individual firm's quantity produced. The market demand curve for this product is

$$\text{Demand:} \quad Q_D = 120 - P$$

where P is the price and Q is the total quantity of the good. Currently, there are 9 firms in the market.

(a) What is each firm's fixed cost? What is its variable cost? Give the equation for average total cost.

(b) At what quantity is the average-total-cost curve at its minimum? What is marginal cost and average total cost at that quantity? Hint: To find the minimum, you'll probably want to use Desmos to graph ATC or use a little calculus.

(c) Give the equation for each firm's supply curve.

(d) Give the equation for the market supply curve for the short run in which the number of firms is fixed.

(e) What is the equilibrium price and quantity for this market in the short run?

(f) In this equilibrium, how much does each firm produce? Calculate each firm's profit or loss. Do firms have an incentive to enter or exit?

(g) In the long run, with free entry and exit, what is the equilibrium price and quantity in this market?

(h) In this long-run equilibrium, how much does each firm produce? How many firms are in the market?

Non-Mankiw Problems

1. Suppose that a market with 10 consumers and 5 producers is perfectly competitive. Each consumer's individual demand is given by $q_D = 40 - 2P$. Each firm's marginal cost is given by $MC = \frac{q_S}{4}$, and fixed costs are \$50.

   (a) What is market equilibrium, and how much does each firm produce?

   (b) What is each firm's variable cost at equilibrium?

   (c) What is each firm's profit at equilibrium?